

A counterexample to Conjecture 00000006542

Ihara poles are not reciprocals of adjacency eigenvalues

Chengzhe Gao

October 3, 2026

The claim being refuted. The English source asserts that “the reciprocals of the poles are the adjacency spectrum.” The Chinese source states the same correspondence. In particular, it implies that if $a \neq 0$ is an Ihara zeta pole, then a^{-1} is an adjacency eigenvalue. We refute this necessary implication for the finite, connected, simple, 2-regular graph C_3 . The source does not exclude cycles or impose minimum degree at least 3. No interpretation of its other clauses about an index, a domain, or coverings is needed for this counterexample.

The standard objects. We use the usual Ihara definition in which a prime is a primitive, closed, backtrackless, tailless directed walk, modulo cyclic rotation. Reversing the direction does not identify two primes. Thus

$$Z_G(u) = \prod_{[P]} (1 - u^{\ell(P)})^{-1}.$$

These conventions are the ones in Horton–Stark–Terras, Section 2, equations (2.1)–(2.2)[1]. Their definition applies to finite connected graphs of fundamental-group rank at least one with no degree-one vertices; C_3 satisfies these conditions. For C_3 the product below is finite, so no convergence theorem or determinant formula is required.

Proposition 1. *The Ihara zeta function of C_3 has a pole at 1, but 1 is not an eigenvalue of its adjacency matrix over $\mathbb{C}$.*

Proof. Use vertices 0, 1, 2 and edges 01, 12, 20. At any directed edge, there is exactly one next edge that is incident to its head and does not immediately reverse it. The six directed edges therefore form the two successor orbits

$$(01, 12, 20), \quad (02, 21, 10).$$

Every backtrackless walk stays in one of these orbits. It closes without a tail if and only if its positive length is divisible by 3. A closed walk of length $3q$, with $q \geq 2$, is the q th power of its first three edges. Conversely a length-three closed walk cannot be a proper power of a shorter reduced closed walk, since the latter would also have positive length divisible by 3. Hence all prime walks have length 3. Cyclic rotation identifies precisely the three starting edges in each displayed orbit, giving exactly two prime classes. Consequently the defining Euler product itself gives

$$Z_{C_3}(u) = \frac{1}{(1 - u^3)^2} = \frac{1}{(u - 1)^2(1 + u + u^2)^2}.$$

The last factor has value 9 at $u = 1$, and the numerator has value 1. Thus there is no cancellation, and $u = 1$ is a pole of exact order 2. Its reciprocal is 1.

The adjacency action on an arbitrary vector $(x, y, z) \in \mathbb{C}^3$ is

$$A(x, y, z) = (y + z, x + z, x + y), \quad A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

If $Av = v$, then $x = y + z$, $y = x + z$, and $z = x + y$. The first two equations imply $x = x + 2z$, so $z = 0$ and $x = y$. The third now gives $2x = 0$, and therefore $x = y = z = 0$. There is no nonzero eigenvector with eigenvalue 1. $\square$

The displayed failure concerns membership in the spectrum, so it does not depend on whether the source meant a set or a multiset of poles and eigenvalues. In the usual theory the relevant direct spectral object is the nonbacktracking directed-edge operator; replacing it by the adjacency operator is the error tested here.

Formalization and coverage. The accompanying `lean/Main.lean` uses only Lean 4.19.0 and `Std`. Its finite computations concern exactly the graph and polynomials above. The classification of walks covers arbitrary positive lengths, not a bounded search.

The graph has vertex type `Fin 3`, with two vertices adjacent exactly when distinct. The six darts have explicit tails, heads, and reverses. Theorems `triangle_connected_regular` and `all_oriented_edges` verify regularity, connectivity, completeness and uniqueness of the dart encoding, and reversal. The predicate `Step` is equality of incident endpoints together with absence of immediate reversal, and `step_is_forced` proves its equivalence to the stated successor permutation.

A `ClosedWalk` is a positive-length periodic dart sequence with `Step` at every pair of consecutive positions. This condition includes the junction between the last and first edge, and thus the usual tailless condition. Every ordinary finite cyclic dart word of every positive length is formally mapped into this representation by `fromFiniteWord`; its periodic extension is indexed modulo that length. Conversely, taking the first declared number of darts of a `ClosedWalk` recovers such a finite cyclic word, because its periodicity identifies the boundary positions. Thus no finite closed walks are omitted. Primitivity explicitly excludes a shorter positive period whose integer repetition number is at least two. Such a period is itself a closed reduced walk, so this is the usual proper-power condition. The symbolic theorem `primitive_iff_length_three` proves the classification for all lengths.

`PrimeCycle` is an actual quotient of primitive walks by rotation, not a two-element type declared to be the primes. The theorem `rotations_iff_direction` proves the equivalence relation and its classification. The maps `cycleCode` and `codeCycle` are proved inverse. In particular, `prime_cycles_complete` and `prime_cycles_nodup` prove that the two displayed representatives are the full index set of the Ihara product, without duplicates. `iharaZeta` multiplies the Euler factors over this complete list. `actual_euler_product` then verifies its numerator and denominator.

Polynomials are finite lists of integer coefficients in ascending degree. Addition and multiplication use coefficient addition and full convolution; evaluation is Horner evaluation. A rational expression is its polynomial numerator and denominator. The predicate `HasPoleAt` uses the exact algebraic criterion

$$D(u) = (u - a)^m Q(u), \quad m > 0, \quad Q(a) \neq 0, \quad N(a) \neq 0.$$

For integer-coefficient rational functions considered over $\mathbb{C}$, this criterion asserts an ordinary meromorphic pole of exact order m : the quotient N/Q is analytic and nonzero near a . The Lean theorem `pole_at_one_order_two` verifies this factorization for the actual Euler product, and `residual_value` verifies $Q(1) = 9$. Its denominator is nonzero since it has value 1 at 0. This finite algebraic pole certificate avoids assuming an unproved analytic continuation or a graph-zeta determinant identity.

The eigenvalue predicate uses all functions `Fin 3` to an arbitrary scalar type, a nonzero coordinate, and the actual matrix action, formed by summing its three entries in each row. The statement `one_not_adjacency_eigenvalue` is proved for every scalar structure `ScalarLaws` with additive associativity, zero, cancellation, no additive 2-torsion, and the usual zero and unit multiplication laws. These are all standard laws of $\mathbb{C}$ (also of any field of characteristic different from 2). Hence the proof applies to arbitrary complex eigenvectors; it is not restricted to integer or rational vectors. The separately supplied integer instance only demonstrates consistency of the small algebraic interface.

Finally `conjecture6542_false` rejects the necessary implication from this actual pole to this actual eigenvalue. The auxiliary script checks graph walks through length 15, the same exact polynomial factorization, and $\det(I - A) = -4$; it is not used to justify the all-length classification or any Lean theorem. The printed axioms of the principal theorems are among `propext`, `Classical.choice`, and `Quot.sound`. There are no placeholders, extra axioms, or unchecked native decision procedures.

References

- [1] M. D. Horton, H. M. Stark, and A. A. Terras, *What are zeta functions of graphs and what are they good for?*, Section 2, author-hosted manuscript, <https://mathweb.ucsd.edu/~aterras/snowbird.pdf>.